Experiment 8

AIM: Develop Inventory Management Module using ADO.NET & CRUD Operations.

Source: InventoryManagementModel Product.cs:-

```
using System;
using System. Collections.
Generic; using System. Linq;
using System. Web;
namespace
InventoryManagementModel. Models



{       public          class
      Product
      { public int ProductId { get; set;
              } public string ProductName { get; set; }
              public int Quantity { get; set; } public decimal
              Price { get; set; }

      }
}
```

ProductController.cs:-

```
using InventoryManagementModel. Models;

using            System.            Data.
SqlClient; using System;
using System. Collections. Generic;
using    System.
Linq;    using
          System. Web; using
          System. Web. Mvc;
          using System. Security. Cryptography. X509Certificates;
 namespace          InventoryManagementModel.
Controllers
{ public class ProductController :
      Controller
      { string cs =@"Data Source=(localdb)\MSSQLLocalDB;
            Initial
Catalog=InventoryDB; Integrated Security=True"; public ActionResult Index()
            {
                  List<Product> products = new
                  List<Product>(); using (SqlConnection conn
                  = new SqlConnection(cs))
                  { string query = "SELECT * FROM Products"; SqlCommand
                  cmd = new SqlCommand(query, conn); conn. Open();

                        SqlDataReader              dr          =          cmd.
                        ExecuteReader(); while (dr. Read())
                        {       products.          Add(new
                              Product
                              {
                                    ProductId = Convert. ToInt32(dr["ProductId"] ),
                              ProductName = dr["ProductName"] . ToString(), Quantity =
                              Convert. ToInt32(dr["Quantity"] ),
                                Price = Convert. ToDecimal(dr["Price"] )
```

```
                    });
                }
            }
            return View(products);
        }
        public ActionResult Create()
        { return View();
        }
        //Crete : save product
        [HttpPost]
        public ActionResult Create(Product p)
        { using (SqlConnection conn = new
            SqlConnection(cs))
            { string query = @"INSERT INTO
                Products
                    (ProductId, ProductName, Quantity, Price)
                 VALUES (@ProductId, @ProductName, @Quantity, @Price)";

                SqlCommand cmd = new SqlCommand(query,
                conn);

                // FIX THE TYPOS HERE (Change @ProuctId to @ProductId,
                etc. ): cmd. Parameters. AddWithValue("@ProductId",
                p.              ProductId);              cmd.
                    Parameters.  AddWithValue("@ProductName",  p.  ProductName);  cmd.
                    Parameters.  AddWithValue("@Quantity",  p.  Quantity);  cmd.  Parameters.
                    AddWithValue("@Price", p. Price);
conn.           Open();              cmd.
                ExecuteNonQuery();
            }
            return RedirectToAction("Index");
        }
        // 1. GET: Product/Edit/5
        [HttpGet]
        public ActionResult Edit(int id)
        {
            Product p = new Product();
            using (SqlConnection conn = new
            SqlConnection(cs))
            { string query = "SELECT * FROM Products WHERE
                                                ProductId=@ProductId";
                        SqlCommand cmd = new SqlCommand(query,
                conn);              cmd.                Parameters.
                AddWithValue("@ProductId",          id);          conn.
                Open();
                SqlDataReader dr = cmd.
                ExecuteReader();
                if (dr. Read())
                {
                    p. ProductId = Convert. ToInt32(dr["ProductId"] );
                p. ProductName = dr["ProductName"] . ToString();
                p. Quantity = Convert. ToInt32(dr["Quantity"] );
                    p.          Price          =          Convert.
                ToDecimal(dr["Price"] ); }
                else { return HttpNotFound(); }
            }
            return View(p);
        }
```

```
        // 2. POST: Product/Edit
        [HttpPost]
        public ActionResult Edit(Product p)
        {
            using (SqlConnection conn = new
            SqlConnection(cs))
            { string query = @"UPDATE Products
                SET ProductName=@ProductName,
                        Quantity=@Quantity,
                        Price=@Price
                    WHERE ProductId=@ProductId";

                    SqlCommand cmd = new SqlCommand(query, conn); cmd.
                Parameters.          AddWithValue("@ProductId",          p.
                ProductId);                cmd.                Parameters.
                AddWithValue("@ProductName",    p.    ProductName);    cmd.
                Parameters.  AddWithValue("@Quantity",  p.  Quantity);  cmd.
                Parameters. AddWithValue("@Price", p. Price); conn. Open(); cmd.
                ExecuteNonQuery();
            }
            return RedirectToAction("Index");
        }
        [HttpPost]
        public ActionResult Delete(int id)
        { using (SqlConnection conn = new
            SqlConnection(cs))
            {
                // FIX TYPO: Change @ProuductId to
                @ProductId
                string query = "DELETE FROM Products WHERE
                ProductId=@ProductId";

                SqlCommand cmd = new SqlCommand(query,
                conn);

                // FIX TYPO: Change @ProuctId to
                @ProductId cmd. Parameters.
                AddWithValue("@ProductId", id);
  conn.         Open();          cmd.
                ExecuteNonQuery();
            }
            return RedirectToAction("Index");
        }
    }
}
```

Create.cshtml:-

```
@modeL InventoryManagementModel. Models. Product
@{
        Layout = null;
}
<h2>ADD NEW PRODUCT</h2>
@using (Html. BeginForm())
{
```

```
        <p>Product Name: </p>
        @Html. TextBoxFor(m => m. ProductName)
        <br      />
        <br />

        <p>Quantity: </p>
        @Html. TextBoxFor(m => m. Quantity)
        <br      />
        <br />

        <p>Price: </p>
        @Html. TextBoxFor(m => m. Price)
        <br      />
        <br />

        <input type="submit" value="SAVE PRODUCT" />
}
<br />
@Html. ActionLink("Back to Inventory", "Index")
```

Index.cshtml:-

```
@model List<InventoryManagementModel. Models. Product>
@{ Layout = null; }
<h2> INVENTORY MANAGEMENT </h2>
<p> @Html. ActionLink("Add new product", "Create")</p>
<table border="1">
    <tr>
        <th>Product ID</th>
        <th>Product Name</th>
        <th>Quantity</th>
        <th>Price</th>
        <th>Actions</th>
    </tr>
    @foreach (var item in Model)
    {
        <tr>
            <td>@item. ProductId</td>
            <td>@item. ProductName</td>
            <td>@item. Quantity</td>
            <td>@item. Price</td>
            <td>
                @Html. ActionLink("Edit", "Edit", new { id = item. ProductId })
                @using (Html. BeginForm("delete", "Product", new { id = item.
                    ProductId }, FormMethod. Post))
                {<input type="submit" value="Delete" />}
            </td>
        </tr>
    }
</table>
```

Edit.cshtml:-

```
@model InventoryManagementModel. Models. Product
@{
        Layout = null;
}
<h2>EDIT PRODUCT</h2>
@using (Html. BeginForm())
{
    @Html. HiddenFor(m => m. ProductId)

    <p>Product Name: </p>
    @Html. TextBoxFor(m => m. ProductName)
    <br />
    <br />

    <p>Quantity: </p>
    @Html. TextBoxFor(m => m. Quantity)
    <br />
    <br />

    <p>Price: </p>
    @Html. TextBoxFor(m => m. Price)
    <br />
    <br />

    <input type="submit" value="UPDATE PRODUCT" />
}
<br />
@Html. ActionLink("Back to Inventory", "Index")
```

RoutConfig. cs

```
 using
System;
using        System.        Collections.
Generic;
using      System.        Linq;
using       System.       Web;
using       System.       Web.
Mvc;      using         System.
Web. Routing;
 namespace
InventoryManagementModel
{ public class
   RouteConfig
   { public static void RegisterRoutes(RouteCollection
      routes)
      { routes. IgnoreRoute("{resource}.
         axd/{*pathInfo}");
  routes. MapRoute(
           name: " Default",
           url: "{controller}/{action}/{id}",
                                defaults: new { controller = " Product", action = " Index", id
                                                                                          =
UrlParameter. Optional }
```

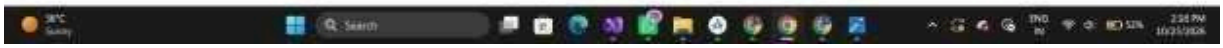
```
            );
        }
    }
}
```

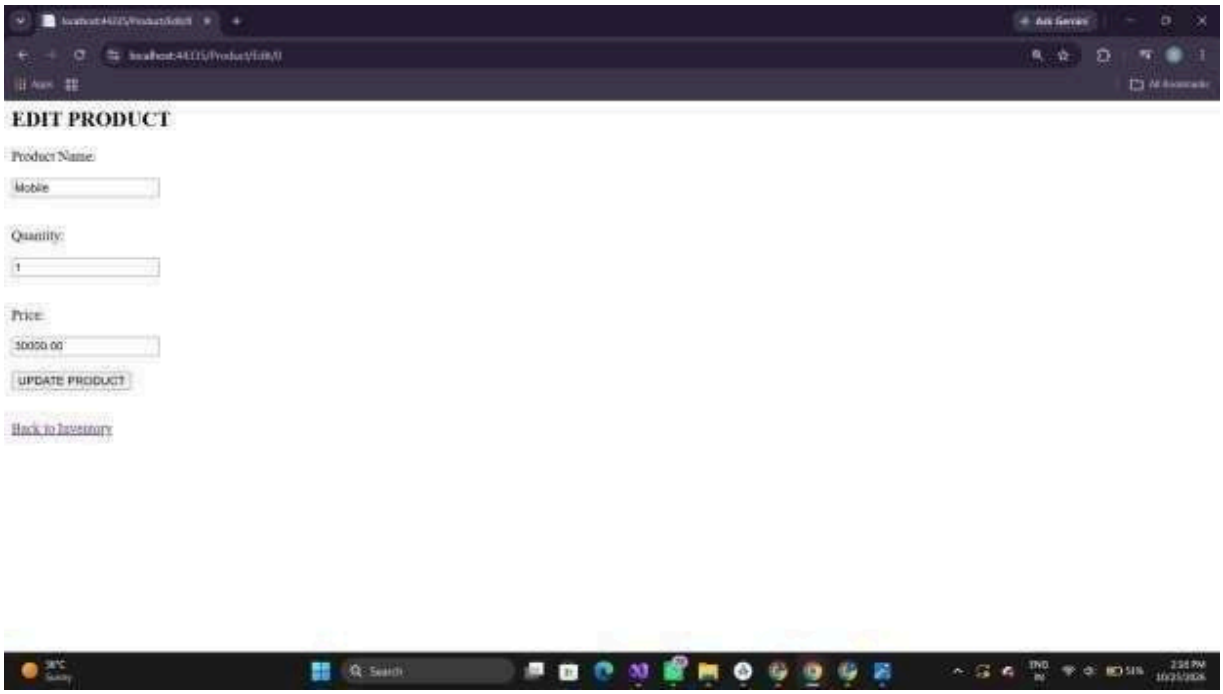
Output: Screenshot:

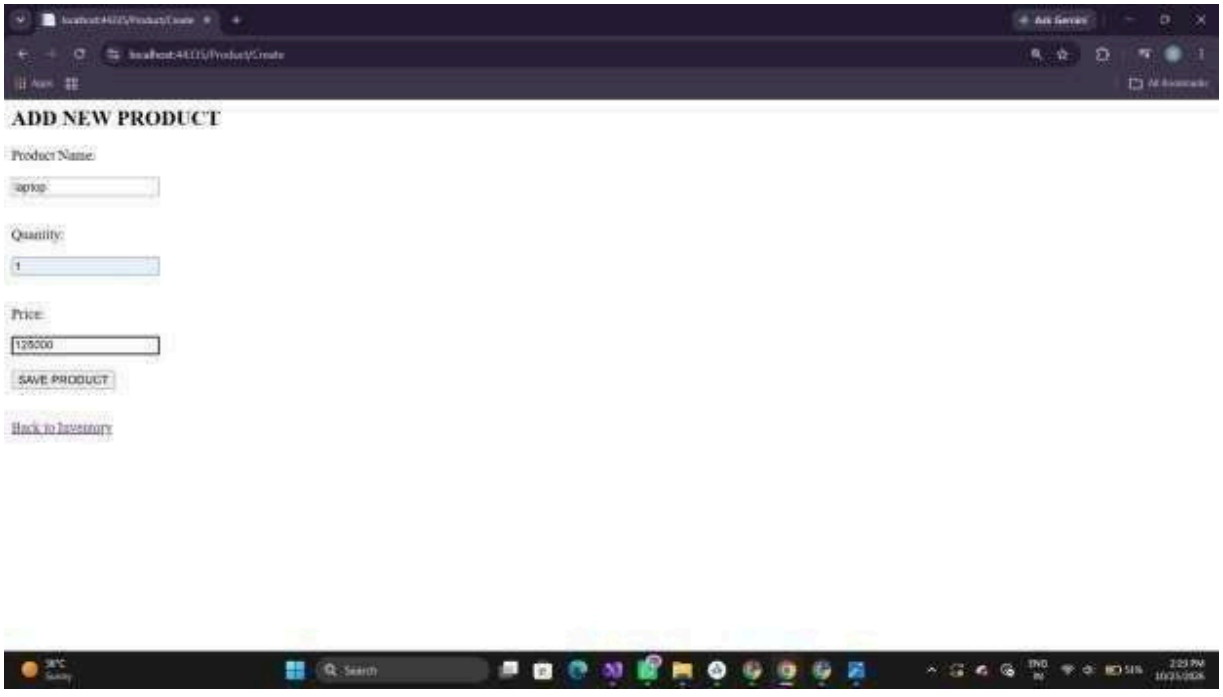


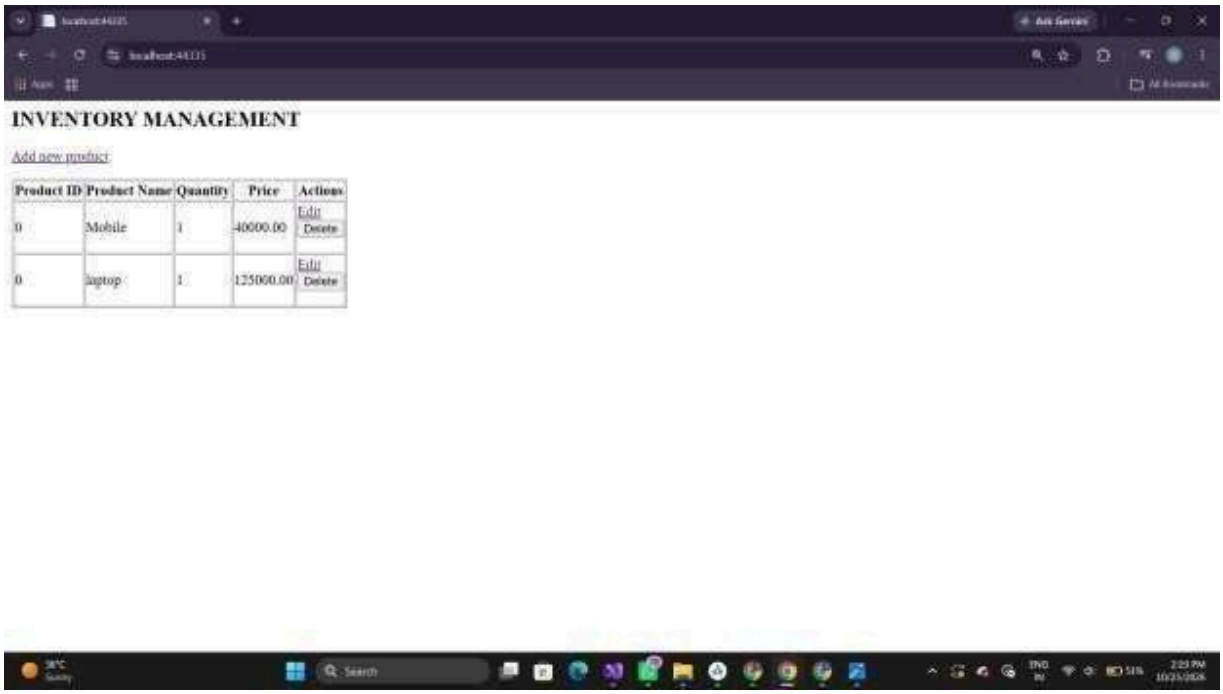







EDIT PRODUCT
Product Name:
Mobile
Quantity:
1
Price:
50000.00
UPDATE PRODUCT
Back to Inventory


ADD NEW PRODUCT
Product Name:
laptop
Quantity:
1
Price:
125000
SAVE PRODUCT
Back to Inventory


INVENTORY MANAGEMENT
Add new product
Product ID Product Name Quantity Price Actions
0 Mobile 1 40000.00 Edit Delete
0 laptop 1 125000.00 Edit Delete